Redrob Candidate Ranker

India Runs: Data & AI Challenge

Team Name: Data Enthusiast

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Problem Statement: Legacy applicant tracking systems rely on rigid keyword matching, missing elite candidates who use different terminology. Our solution aims to solve this by building a highly intelligent, 5-stage semantic AI with advanced psychological profiling.

Solution Overview

What is your proposed solution?

An elite, "God-Tier" Omni-Architecture utilizing a Two-Stage NLP pipeline (Bi-Encoder + Cross-Encoder) combined with psychological profiling, heuristic behavioral extraction, and live dynamic Streamlit dashboarding.

What differentiates your approach?

1. **Psychological Synthesis:** We automatically synthesize a proxy for the candidate's "Big Five" (OCEAN) personality traits using open-source heuristic formulas.
2. **"Blind Audition" Anti-Bias Mode:** Our UI allows recruiters to toggle a mode that instantly anonymizes candidate names to enforce 100% merit-based hiring.
3. **Omega Predictors:** We inject 7 predictive layers including Estimated Salary, Time-to-Hire, Flight Risk, Career Trajectory, and Competitor Poach Value.

JD Understanding & Candidate Evaluation

What are the key requirements extracted?

Core AI competencies (embeddings, retrieval, python), product-company experience (vs pure service), and minimum 3 years of experience.

Which candidate signals are most important?

We evaluate candidates on deep psychological dimensions:

1. **Semantic Match:** Deep contextual similarity to the JD.
2. **Career Stability & Promotion Velocity:** Rewarding high-performers.
3. **Leadership Bias:** Tracking their use of active vs passive verbs.
4. **Modesty (Dunning-Kruger):** Penalizing juniors who claim 20+ "Expert" skills.
5. **Resume Fluff Detection:** Flagging those who overuse empty buzzwords like "synergy".

Ranking Methodology

How does your system retrieve, score, and rank?

1. **Hard Filter:** Drops candidates with `<3` YoE or honeypot traps.
2. **Behavioral Scoring:** Factors in response rates, notice periods, and learner indices.
3. **Bi-Encoder Re-Ranking:** Top 1,000 scored via Cosine Similarity (`all-MiniLM-L6-v2`).
4. **Cross-Encoder Re-Ranking:** Top 200 run through a deep-interaction Microsoft Cross-Encoder.
5. **Reciprocal Rank Fusion (RRF):** Perfect mathematical aggregation of discrete ranks.
6. **Percentile Distribution:** Ranks are contextualized into Peer Percentile Badges (e.g., *Top 1% Stability*).

Explainability & Data Validation

How are ranking decisions explained?

- **Generative AI Summaries:** An instant, rule-based NLP engine writes a 3-paragraph executive summary explaining exactly why the candidate is a perfect fit.
- **Dynamic Interview Questions:** The AI generates 3 custom interview questions per candidate based on their exact red flags and technical claims.
- **Visual Graphs:** Our Streamlit UI provides 3D Plotly Radar Charts and interactive Graphviz Knowledge Graphs to visually explain the candidate's brain.

How do you prevent hallucinations?

No Generative LLM is used at runtime. All reasoning and summaries are synthesized deterministically from verified metadata, ensuring 100% factual accuracy.

End-to-End Workflow

1. **Ingest:** Stream massive `candidates.jsonl` files iteratively.
2. **Two-Stage NLP:** Run the Bi-Encoder -> Cross-Encoder dual architecture.
3. **Psychological Profiling:** Synthesize the Big Five personality traits, Leadership score, and Red Flag detection.
4. **Omega Predictions:** Calculate Salary, Time-to-Hire, and Retention Probabilities.
5. **Generate Output:** Instantly write the Executive Summary, Draft Offer Letters, and export `team_PJ2001_IND.csv`.
6. **Visualization:** Our custom Hugging Face UI dynamically renders the candidate's complete psychological footprint.

System Architecture

- **Data Layer:** Memory-efficient JSONL streaming parser.
- **Retrieval Engine:** Custom BM25 with Latent Skill Inference.
- **NLP Engine:** HuggingFace `sentence-transformers` (`all-MiniLM` + `ms-marco`).
- **Aggregation Layer:** Reciprocal Rank Fusion (RRF).
- **Presentation Layer:** A stunning Streamlit UI built with `plotly` (Radar Charts) and `graphviz` (Skill Knowledge Networks).

(Everything runs 100% locally. No closed-source external APIs are used, ensuring absolute data privacy and infinite scalability.)

Results & Performance

What results demonstrate ranking quality?

The system successfully identified highly-qualified Senior Engineers who possessed the required skills but suffered from "resume modesty" by automatically inferring their latent skills and rewarding their high career velocity, while penalizing Dunning-Kruger inflators.

How does it meet compute constraints?

- **Scale:** Successfully evaluated the entire 100,000 candidate dataset.
- **Speed:** Even while computing Cross-Encoders, Temporal Consistency, Graph generation, and the 7 Omega Predictors, the total end-to-end runtime is **~8 to 23 seconds** on standard CPU.
- **Memory:** Stays safely below limits using efficient data structures.

Technologies Used

- **Python (3.11+):** Core pipeline architecture.
- **Sentence-Transformers:** For free, open-source, CPU-optimized NLP models.
- **Plotly & Graphviz:** For generating 3D interactive Radar Charts and spider-web Knowledge Graphs natively in the UI.
- **Scikit-Learn:** For rapid matrix calculations.
- **Streamlit:** For hosting the interactive "X-Ray Vision" dashboard.
- **Hugging Face Spaces:** Used to host the live interactive Sandbox environment.

Submission Assets

- **Live Sandbox Demo:** https://huggingface.co/spaces/praasukjain2001/redrob-ranker
- **GitHub Repository:** https://github.com/PJ2001-IND/redrob-ranker
- **Output File:** `team_PJ2001_IND.csv` (and `.xlsx`)
- **Metadata:** `submission_metadata.yaml`

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